Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[14EC33]**

THIRD SEMESTER B.TECH EXAMINATION DECEMBER-2015

ANALOG ELECTRONIC CIRCUITS

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full questions, choosing at least **TWO** from each part.

PART - A

1. a. The reverse saturation current of silicon diode is 3 nA at27^0 Find the reverse saturation current flowing through the diode at 82°c. Also find the current flowing through the diode at 82°c if the forward voltage drop across the diode is 0.7 volts. Take n=2. **4**

   b. Draw the output characteristics of JFET connected in CS mode and also write the relation between drain current v/s V_{GS}. Indicate the region and condition under which JFET can be used as an amplifier. **6**

   c. Draw the PNP transistor circuit connected in CC mode and also the output characteristics. **4**

2. a. What is the need for biasing in transistor? **2**

   b. Explain the analysis of voltage divider bias network using exact analysis **6**

   c. Derive an expression to fond stability factor S_B in a fixed bias circuit. Also find the change in collector current when $\beta_{25}°c=50$ changes to $\beta_{75}°c=100$. Given $I_{c1}=3mA$ **6**

3. a. Draw the frequency response of an single stage RC coupled amplifier and also explain the effect of capacitance on low frequency, mid frequency and high frequency. **5**

   b. For the circuit shown in fig 3(b) find the, midband voltage gain Av and Av_s, lower cut off frequency due to C_S, lower cut off frequency duw to C_C, lower cut off frequency duw to C_E, over all lower cut off frequency for the transistor B=100. And $r_o=\infty$ **9**

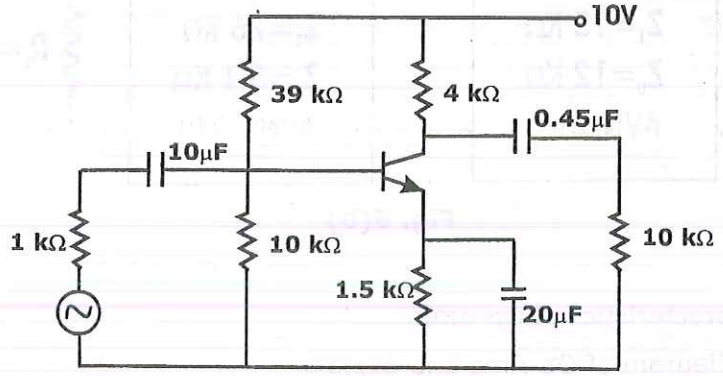

Fig. 3(b)

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4. a. A full wave bridge rectifier with 220 volts rms sinusoidal input has a load resistor of 10k Ω. Assuming the ideal diodes are used calculate

   i. DC output voltage and current

   ii. Peak output voltage and current

b. For the circuit shown in big Fig.4 (b) draw the output wave form use v_γ=0.7 for Si. **6**

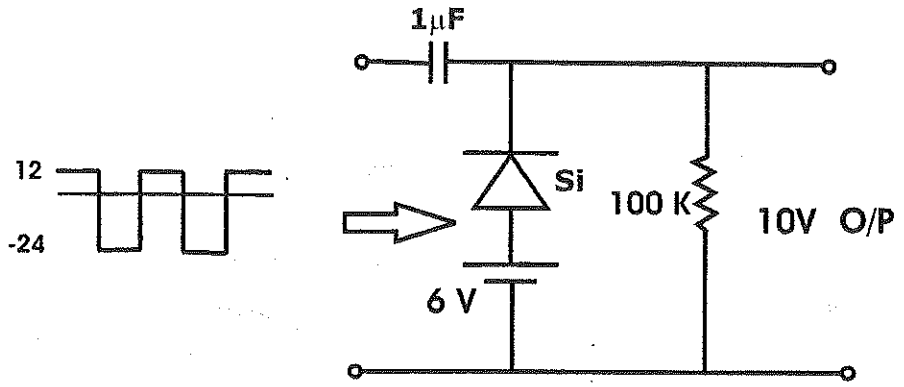

Fig. 4(b)

c Find the maximum conversion efficiency in class B power amplifier. **4**

PART- B

5. a. Write the advantages of negative feedback. **4**

b. Draw the circuit diagram of voltage series feedback amplifier and also find an expression for input resistance and output resistance. **6**

c In an RC phase shift oscillator the collector resistance R_c=5 kΩ and the value of resistance R=3.3 kΩ in the feedback path. Find the range of variable capacitance to be used in the feedback path to vary the frequency from 100 Hertz to 20 KHz. **4**

6. a. What is cascading? Find an expression to calculate the total voltage gain when 'n' amplifiers is cascaded. **7**

b. For the cascaded arrangement shown in Fig. 6(b) calculate the voltage gain of each stage, the total gain A_V and A_{VS} of the system. **7**

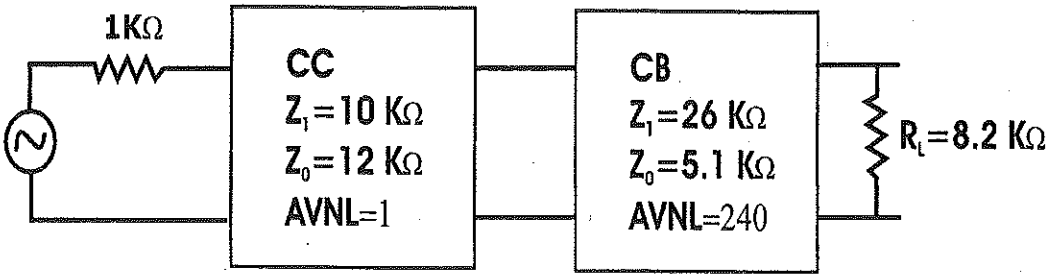

Fig. 6(b)

7. a. List the ideal characteristics of Op-Amp. **4**

b. Draw the block diagram of Op-Amp and explain. **6**

c. Derive an expression to calculate the voltage gain of an inverting amplifier. **4**

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8. a. What is three terminal fixed voltage regulator? Explain the working operation with schematic diagram. **7**

b. An LM 317 regulator is to provide 6 volts output from a 15 volts supply determine the value of resistance R2 in big 8(b) given $I_1 = 1$ mA. **7**

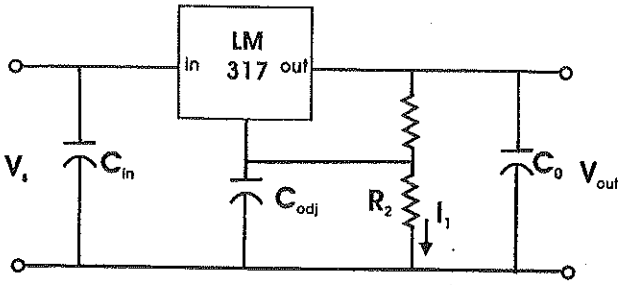

Fig. 8(b)